Chapter 6
Induction

The Mechanics of Proof | CSC302 lecture notes

Based on Heather Macbeth's textbook

6.1 Induction: an infinite chain of implications

For a proposition $Q(n)$ indexed by $n \in \mathbb{N} = \{0, 1, 2, \ldots\}$, prove

$$Q(0), \qquad \forall k \in \mathbb{N}, \quad Q(k) \Longrightarrow Q(k+1).$$

These two facts establish $Q(n)$ for every natural number n.

Worked example: $2^n \geq n + 1$ for every $n \in \mathbb{N}$

Base: $2^0 = 1 = 0 + 1$. Fix $k \in \mathbb{N}$ and assume $2^k \geq k + 1$. Then

$$2^{k+1} = 2 \cdot 2^k \geq 2(k+1) = (k+2) + k \geq k + 2.$$

The last inequality uses $k \geq 0$; the first uses the hypothesis at k.

The induction hypothesis is available at k, while the goal concerns $k + 1$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.1, Example 6.1.1.

Lean: identify the base and successor goals

Here is a closely related claim, $n + 1 \leq 3^n$, over $\mathbb{N}$.

```
example (n : Nat) : n + 1 <= 3 ^ n := by
  simple_induction n with k IH
  . numbers
  . calc
      k + 1 + 1 <= 3 * (k + 1) := by addarith
      _ <= 3 * 3 ^ k := by rel [IH]
      _ = 3 ^ (k + 1) := by ring
```

- `simple_induction` gives the base goal and a successor goal with `IH`.
- `rel [IH]` transports an inequality through multiplication by 3.
- `addarith` handles linear arithmetic; `ring` handles the identity.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.1, Example 6.1.1.

Powers preserve congruence: prove the general rule

Let $a, b, m \in \mathbb{Z}$ with $a \equiv b \pmod{m}$. We prove $a^n \equiv b^n \pmod{m}$ for every $n \in \mathbb{N}$.
Choose $v \in \mathbb{Z}$ with $a - b = mv$.
Base: $a^0 - b^0 = 1 - 1 = m \cdot 0$.
Step: fix $k \in \mathbb{N}$ and assume $a^k - b^k = mu$ for some $u \in \mathbb{Z}$. Then

$$\begin{aligned} a^{k+1} - b^{k+1} &= a(a^k - b^k) + b^k(a - b) \\ &= a(mu) + b^k(mv) \\ &= m(au + b^k v). \end{aligned}$$

The new witness $au + b^k v$ is an integer. Induction completes the proof. No division by m is used, so the argument also covers $m = 0$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.1, Example 6.1.3; general rule used in Chapter 3.

Starting at a threshold

"For all sufficiently large n" means

$$\exists N \in \mathbb{N}\, \forall n \in \mathbb{N}, \quad N \leq n \Longrightarrow Q(n).$$

For $Q(n) : 2^n \geq n^2$, choose $N = 4$. The base is $16 = 4^2$. For $k \geq 4$, assume $2^k \geq k^2$. Since

$$2k^2 - (k+1)^2 = k(k-2) - 1 \geq 4 \cdot 2 - 1 = 7,$$

we get $2^{k+1} \geq 2k^2 \geq (k+1)^2$.

Lean setup pattern for a goal with `hn : 4 <= n`:

```
induction_from_starting_point n, hn with k hk IH
```

The step now includes `hk : 4 <= k`; the base is at 4, not 0.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.1, Examples 6.1.5–6.1.6.

Practice 6.1: a disjunctive invariant

Prove, for every $n \in \mathbb{N}$,

$$5^n \equiv 1 \pmod 8 \quad \text{or} \quad 5^n \equiv 5 \pmod 8.$$

All powers and congruences here are interpreted in $\mathbb{Z}$.

1. State the base case at $n = 0$.
2. Write the exact induction hypothesis at k.
3. Split that hypothesis into two cases. Which alternative follows at $k + 1$?

In Lean, a disjunctive hypothesis can be split using `obtain h1 | h5 := IH`; choose the goal's alternative using `left` or `right`.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.1.7.

Solution 6.1: multiplication swaps the alternatives

Base: $5^0 = 1$, so choose the first alternative.
Fix $k \in \mathbb{N}$ and assume $5^k \equiv 1$ or $5^k \equiv 5 \pmod 8$.

- If $5^k \equiv 1$, then
$$5^{k+1} = 5 \cdot 5^k \equiv 5 \cdot 1 = 5 \pmod 8.$$
- If $5^k \equiv 5$, then
$$5^{k+1} = 5 \cdot 5^k \equiv 25 \equiv 1 \pmod 8,$$
since $25 - 1 = 8 \cdot 3$.

Thus the same disjunction holds at $k + 1$. Induction completes the proof.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.1.7.

6.2 Recursion defines; induction proves

A sequence with values in a set S is a function $\mathbb{N} \to S$. A recurrence specifies initial values and rules for later values.
For example, define $b : \mathbb{N} \to \mathbb{Z}$ by

$$b_0 = 3, \qquad b_{n+1} = b_n^2 - 2 \quad (n \in \mathbb{N}).$$

Its first terms are $3, 7, 47, 2207$. Each call uses $n < n + 1$.

Invariant: every b_n is odd

Base: $b_0 = 2 \cdot 1 + 1$. For $k \in \mathbb{N}$, assume $b_k = 2t + 1$ for some $t \in \mathbb{Z}$. Then

$$b_{k+1} = (2t + 1)^2 - 2 = 2(2t^2 + 2t - 1) + 1.$$

The displayed integer $2t^2 + 2t - 1$ is the new oddness witness.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.2, Example 6.2.1.

A closed formula from a shifted recurrence

Let $x_0 = 5$ and $x_{n+1} = 2x_n - 1$, with $x : \mathbb{N} \to \mathbb{Z}$. Subtract the fixed point 1: $x_{n+1} - 1 = 2(x_n - 1)$.

```
def x : Nat -> Int
  | 0 => 5
  | k + 1 => 2 * x k - 1

example (n : Nat) : x n - 1 = 4 * 2 ^ n := by
  simple_induction n with k IH
  . rw [x]; numbers
  . calc
      x (k + 1) - 1 = 2 * (x k - 1) := by rw [x]; ring
      _ = 2 * (4 * 2 ^ k) := by rw [IH]
      _ = 4 * 2 ^ (k + 1) := by ring
```

Hence $x_n = 2^{n+2} + 1$, and $x_n \equiv 1 \pmod 4$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.2, Examples 6.2.2–6.2.3.

A recursively defined sum

Define $A : \mathbb{N} \to \mathbb{Q}$ by $A_0 = 0$ and $A_{n+1} = A_n + (n+1)$ for $n \in \mathbb{N}$. Thus $A_n = 1 + 2 + \cdots + n$, with the empty sum equal to 0.

Claim: $A_n = \dfrac{n(n+1)}{2}$ for all $n \in \mathbb{N}$

Base: $A_0 = 0 = 0(0+1)/2$. For $k \in \mathbb{N}$, assume $A_k = k(k+1)/2$. Then

$$\begin{aligned} A_{k+1} &= A_k + (k+1) \\ &= \frac{k(k+1) + 2(k+1)}{2} = \frac{(k+1)(k+2)}{2}. \end{aligned}$$

Lean connection: `rw [A]` exposes the recurrence, `rw [IH]` replaces the smaller sum, and `ring` finishes in $\mathbb{Q}$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.2, Example 6.2.4.

Factorials: keep the divisor universally quantified

Define $0! = 1$ and $(n+1)! = (n+1)n!$ for $n \in \mathbb{N}$. Induction also gives $n! \geq 1$: the base is 1, and both factors in $(k+1)k!$ are at least 1.

Claim: $\forall n, d \in \mathbb{N},\ 1 \leq d \leq n \implies d \mid n!$

Induct on n, keeping the claim about *every* such d. At $n = 0$, no d satisfies $1 \leq d \leq 0$. At $n = k+1$, let $1 \leq d \leq k+1$.

- If $d = k+1$, then $(k+1)! = d\,k!$.
- Otherwise $d \leq k$. The IH gives $k! = dt$ for some $t \in \mathbb{N}$, so $(k+1)! = d((k+1)t)$.

In Lean, introduce d *after* `simple_induction n`.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.2, Example 6.2.5.

Practice 6.2: formulas and bounds

1. Define $c : \mathbb{N} \to \mathbb{Z}$ by

$$c_0 = 7, \qquad c_{n+1} = 3c_n - 10 \quad (n \in \mathbb{N}).$$

   Prove $c_n = 2 \cdot 3^n + 5$ for all $n \in \mathbb{N}$. What fixed point should you subtract from the recurrence?

2. Prove $(n+1)! \geq 2^n$ for every $n \in \mathbb{N}$. Write the successor factorial so that the induction hypothesis applies.

For each proof, name the base case and state the IH before calculating.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.2.7; Example 6.2.6.

Solution 6.2: expose one recursive layer

1. Closed formula. The fixed point is 5, since $3 \cdot 5 - 10 = 5$. Base: $7 = 2 \cdot 3^0 + 5$. For $k \in \mathbb{N}$, assuming $c_k = 2 \cdot 3^k + 5$ gives

$$c_{k+1} = 3(2 \cdot 3^k + 5) - 10 = 2 \cdot 3^{k+1} + 5.$$

2. Factorial bound. Base: $1! = 1 = 2^0$. For $k \in \mathbb{N}$, assuming $(k+1)! \geq 2^k$ gives

$$(k+2)! = (k+2)(k+1)! \geq (k+2)2^k \geq 2 \cdot 2^k = 2^{k+1}.$$

The second inequality uses $k + 2 \geq 2$ and $2^k \geq 0$.
Both inductions follow the decreasing index in the recursive definition.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.2.7; Example 6.2.6.

6.3 Two base cases, two induction hypotheses

To establish $Q(n)$ for all $n \in \mathbb{N}$, it suffices to prove

$$Q(0), \quad Q(1), \quad \forall k \in \mathbb{N},\ (Q(k) \wedge Q(k+1)) \Longrightarrow Q(k+2).$$

Let $a : \mathbb{N} \to \mathbb{Z}$ satisfy $a_0 = 2$, $a_1 = 1$, $a_{n+2} = a_{n+1} + 2a_n$. Claim: $a_n = 2^n + (-1)^n$.
Bases: $2 = 1 + 1$ and $1 = 2 - 1$. **Step:** for $k \in \mathbb{N}$, assume the formula at both k and $k+1$.

$$\begin{aligned} a_{k+2} &= (2^{k+1} + (-1)^{k+1}) + 2(2^k + (-1)^k) \\ &= 4 \cdot 2^k + (-1)^k = 2^{k+2} + (-1)^{k+2}. \end{aligned}$$

Both recursive arguments, k and $k+1$, are smaller than $k+2$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.3, Example 6.3.1.

Fibonacci: an exponential upper bound

Define $F : \mathbb{N} \to \mathbb{Z}$ by

$$F_0 = F_1 = 1, \qquad F_{n+2} = F_{n+1} + F_n \quad (n \in \mathbb{N}).$$

The first terms are $1, 1, 2, 3, 5, 8, 13$.

Claim: $F_n \leq 2^n$ for all $n \in \mathbb{N}$

Bases: $F_0 = 1 \leq 1$ and $F_1 = 1 \leq 2$. For $k \in \mathbb{N}$, assume $F_k \leq 2^k$ and $F_{k+1} \leq 2^{k+1}$. Then

$$F_{k+2} \leq 2^{k+1} + 2^k = 3 \cdot 2^k \leq 4 \cdot 2^k = 2^{k+2}.$$

The bound at $k+1$ alone would leave F_k uncontrolled.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.3, Example 6.3.3.

Sometimes the induction hypothesis needs more information

For the sequence $a_0 = 2$, $a_1 = 1$, $a_{n+2} = a_{n+1} + 2a_n$, prove $a_n \equiv 1$ or $5 \pmod 6$ for $n \geq 1$.
Use the stronger assertion that the pair (a_n, a_{n+1}) is congruent to

$$(1,5) \quad \text{or} \quad (5,1) \pmod 6.$$

Base $n = 1$: $(a_1, a_2) = (1,5)$. For $k \geq 1$, assume this assertion at k. The recurrence gives

$$(1,5) \longmapsto (5, 5 + 2 \cdot 1) \equiv (5,1),$$

$$(5,1) \longmapsto (1, 1 + 2 \cdot 5) \equiv (1,5).$$

Ordinary induction now works on the *pair assertion*; its first component implies the original claim. This excludes the unwanted pairs $(1,1)$ and $(5,5)$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.3, Example 6.3.2.

Two-step induction from a later starting point

Define $d : \mathbb{N} \to \mathbb{Z}$ by $d_0 = 3$, $d_1 = 1$, $d_{n+2} = 3d_{n+1} + 5d_n$. To prove $d_n \geq 4^n$ for $n \geq 4$, check *two* bases:

$$d_4 = 267 \geq 256, \qquad d_5 = 1096 \geq 1024.$$

For $k \geq 4$, the hypotheses at $k, k+1$ give

$$d_{k+2} \geq 3 \cdot 4^{k+1} + 5 \cdot 4^k = 17 \cdot 4^k \geq 4^{k+2}.$$

Lean setup patterns:

```
two_step_induction n with k IH0 IH1
-- Or, with hn : 4 <= n:
two_step_induction_from_starting_point n, hn with k hk IH0 IH1
```

Here `IH0` concerns k and `IH1` concerns $k+1$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.3, Example 6.3.5.

Practice 6.3: a second-order recurrence

Define $t : \mathbb{N} \to \mathbb{Z}$ by

$$t_0 = 5, \qquad t_1 = 7, \qquad t_{n+2} = 2t_{n+1} - t_n \quad (n \in \mathbb{N}).$$

Prove $t_n = 2n + 5$ for every $n \in \mathbb{N}$.

1. Check both bases.
2. State separate hypotheses for t_k and t_{k+1}.
3. Substitute both formulas into the recurrence.
4. Explain why checking only t_0 cannot determine the whole sequence.

When expressing the formula in Lean, use an integer cast of n.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.3.6.

Solution 6.3: the two initial values matter

Bases: $t_0 = 5 = 2 \cdot 0 + 5$ and $t_1 = 7 = 2 \cdot 1 + 5$.
Let $k \in \mathbb{N}$. Assume $t_k = 2k + 5$ and $t_{k+1} = 2(k+1) + 5$. Then

$$\begin{aligned} t_{k+2} &= 2(2(k+1) + 5) - (2k + 5) \\ &= 2k + 9 = 2(k+2) + 5. \end{aligned}$$

The recurrence has the same first difference at each step: $t_{n+2} - t_{n+1} = t_{n+1} - t_n$. Changing t_1 to 8 while keeping $t_0 = 5$ would instead produce $5, 8, 11, 14, \ldots$. The value at 0 does not fix the initial difference.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.3.6.

6.4 Strong induction: use any smaller input

For $Q : \mathbb{N} \to \mathrm{Prop}$, prove for every $n \in \mathbb{N}$:

$$(\forall m \in \mathbb{N},\ m < n \Longrightarrow Q(m)) \Longrightarrow Q(n).$$

Then $Q(n)$ holds for all n. At $n = 0$ there are no smaller inputs.

Worked proof: every $n \geq 2$ has a prime divisor

IH: every m with $2 \leq m < n$ has a prime divisor. If n is prime, use $p = n$, since $n = n \cdot 1$. Otherwise, the proper-factor lemma gives $n = mx$ with $2 \leq m < n$. The IH gives $m = py$ for a prime p, hence $n = p(yx)$.

The least relevant input $n = 2$ is prime. The recursive branch uses $m < n$, which need not be $m = n - 1$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.4, Examples 6.4.1–6.4.2.

Lean: a theorem can call itself on smaller inputs

Using the preceding definition of F:

```
theorem fib_upper (n : Nat) : F n <= (2 : Int) ^ n := by
  match n with
  | 0 => rw [F]; numbers
  | 1 => rw [F]; numbers
  | k + 2 =>
      have h0 := fib_upper k
      have h1 := fib_upper (k + 1)
      calc
        F (k + 2) = F (k + 1) + F k := by rw [F]
        _ <= 2 ^ (k + 1) + 2 ^ k := by rel [h0, h1]
        _ = 3 * 2 ^ k := by ring
        _ <= 3 * 2 ^ k + 2 ^ k := by extra
        _ = 2 ^ (k + 2) := by ring
```

Both calls decrease the natural-number input. A call at $k + 2$ would be circular.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.4, Example 6.4.1.

Practice 6.4: remove all factors of two

Show that every $n \in \mathbb{N}$ with $n > 0$ admits a representation

$$n = 2^a x, \qquad a, x \in \mathbb{N}, \qquad x \text{ odd}.$$

An odd natural number has the form $2u + 1$ with $u \in \mathbb{N}$.

1. What witnesses work when n is odd?
2. If $n = 2m$ is positive, prove $0 < m < n$ before invoking the IH.
3. Convert a representation of m into one for n.

The measure is the positive input n; the recursive call is at $m = n/2$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.4.3.

Solution 6.4: split on parity

Use strong induction. At a positive n, assume the statement for every m with $0 < m < n$.

- If n is odd, choose $a = 0$, $x = n$: $n = 2^0 n$.
- If n is even, write $n = 2m$. Positivity gives $m \geq 1$; then $m < 2m = n$. By the IH, $m = 2^a x$ for some odd x. Thus

$$n = 2m = 2 \cdot 2^a x = 2^{a+1} x.$$

  Use witnesses $a + 1$ and x.

At $n = 1$ the odd branch closes directly. At $n = 0$ the premise $n > 0$ is false, so no recursive call or witnesses are required.

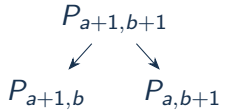
Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.4.3.

6.5 Pascal's triangle in two coordinates

Define $P : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ by

$$P(a, 0) = P(0, b) = 1, \qquad P(a+1, b+1) = P(a+1, b) + P(a, b+1).$$

```
def pascal : Nat -> Nat -> Nat
  | a, 0 => 1
  | 0, b + 1 => 1
  | a + 1, b + 1 =>
      pascal (a + 1) b + pascal a (b + 1)
termination_by _ a b => a + b
```



Each dependency has coordinate sum $a + b + 1$, less than $a + b + 2$.

For example: $P(1, 1) = 2$, $P(1, 2) = 3$, $P(2, 2) = 3 + 3 = 6$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.5, Definition 6.5.1.

The factorial identity: choose the exact invariant

For all $a, b \in \mathbb{N}$,

$$\boxed{P(a,b)\, a!\, b! = (a+b)!}.$$

Induct on $a+b$, allowing the IH at *every* pair with smaller sum.

- If $b = 0$, then $P(a,0)a!0! = 1 \cdot a! \cdot 1 = (a+0)!$.
- If $a = 0$, then $P(0,b)0!b! = 1 \cdot 1 \cdot b! = (0+b)!$.

For the interior pair $(a+1, b+1)$, the two needed instances are

$$P(a+1,b)(a+1)!b! = (a+b+1)!,$$

$$P(a,b+1)a!(b+1)! = (a+b+1)!.$$

Each is justified by a decrease of one in the coordinate sum.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.5, Example 6.5.3.

The factorial identity: align the two hypotheses

Write $U = P(a+1, b)$, $V = P(a, b+1)$, and $s = a + b + 1$. The IH says $U(a+1)!b! = s!$ and $Va!(b+1)! = s!$. Then

$$\begin{aligned} P(a+1,b+1)(a+1)!(b+1)! &= (U+V)(a+1)!(b+1)! \\ &= (b+1)\big[U(a+1)!b!\big] \\ &\quad + (a+1)\big[Va!(b+1)!\big] \\ &= (b+1)s! + (a+1)s! = (s+1)!. \end{aligned}$$

Since $a!b! \geq 1$, the identity implies $P(a,b) \leq (a+b)!$. Since factorials are nonzero, division in $\mathbb{Q}$ gives $P(a,b) = (a+b)!/(a!b!)$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.5, Examples 6.5.2–6.5.3.

Practice 6.5: an edge and a symmetry

For all natural numbers a, b, prove:

1. $P(a, 1) = a + 1$, using ordinary induction on a.
2. $P(a, b) = P(b, a)$, using induction on $a + b$.

For the symmetry proof, state the IH for arbitrary pairs (x, y) with $x + y < a + b$. Handle the two boundary directions before the interior.

Lean connection: unfold `pascal` with `rw [pascal]`. For the interior symmetry case, recursively use the symmetry theorem at $(a + 1, b)$ and $(a, b + 1)$, then use commutativity of addition.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.5.4.

Solution 6.5: respect both coordinates

Edge. Base: $P(0,1) = 1$. Assume $P(k,1) = k+1$. Then

$$P(k+1,1) = P(k+1,0) + P(k,1) = 1 + (k+1) = k+2.$$

Symmetry. If either coordinate is zero, both sides are 1. For $(a+1, b+1)$, assume symmetry for all smaller sums. Then

$$\begin{aligned} P(a+1,b+1) &= P(a+1,b) + P(a,b+1) \\ &= P(b,a+1) + P(b+1,a) \\ &= P(b+1,a) + P(b,a+1) \\ &= P(b+1,a+1). \end{aligned}$$

Both IH calls have sum $a+b+1 < a+b+2$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.5.4.

6.6 Recursive division on all integers

Let $M(n, d) = \operatorname{mod}(n, d)$ and $D(n, d) = \operatorname{div}(n, d)$, both functions $\mathbb{Z}^2 \to \mathbb{Z}$. Use the first applicable row:

Condition	$M(n, d)$	$D(n, d)$
$nd < 0$	$M(n + d, d)$	$D(n + d, d) - 1$
$0 < d(n - d)$	$M(n - d, d)$	$D(n - d, d) + 1$
$n = d$	0	1
otherwise	n	0

For $d > 0$, these respectively address $n < 0$, $n > d$, $n = d$, and $0 \leq n < d$.
The definitions also cover $d \leq 0$. For example, $M(n, 0) = n$, while $D(0, 0) = 1$ and $D(n, 0) = 0$ for $n \neq 0$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.6, Definition 6.6.1.

Why recursive division terminates

Use the nonnegative integer measure $\mu(n, d) = |2n - d|$. The value of n itself may increase or be negative.

- If $nd < 0$, the call is at $(n + d, d)$. The difference of squared measures is

$$(2n + d)^2 - (2n - d)^2 = 8nd < 0.$$

- If $d(n - d) > 0$, the call is at $(n - d, d)$, and

$$(2n - 3d)^2 - (2n - d)^2 = -8d(n - d) < 0.$$

In both cases, the new absolute value is strictly smaller. The terminal rows make no calls. Example: dividing -7 by 3 visits $-7, -4, -1, 2$; the measures are $17, 11, 5, 1$. Thus $M(-7, 3) = 2$ and $D(-7, 3) = -3$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.6, Definition 6.6.1.

Correctness, part I: reconstruct the dividend

Claim: $M(n,d) + dD(n,d) = n$ for all $n, d \in \mathbb{Z}$. Induct on $|2n - d|$. The IH is this identity at every (m, c) with $|2m - c| < |2n - d|$.

- If $nd < 0$, put $M' = M(n + d, d)$ and $D' = D(n + d, d)$. The IH says $M' + dD' = n + d$, so

$$M(n,d) + dD(n,d) = M' + d(D' - 1) = (n + d) - d = n.$$

- If $d(n - d) > 0$, use the IH $M' + dD' = n - d$ at $(n - d, d)$:

$$M(n,d) + dD(n,d) = M' + d(D' + 1) = (n - d) + d = n.$$

- The $n = d$ row gives $0 + d \cdot 1 = n$; the final row gives $n + d \cdot 0 = n$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.6, Example 6.6.2.

Correctness, part II: bound the remainder

Claim: for $n, d \in \mathbb{Z}$ with $d > 0$,

$$0 \leq M(n, d) < d.$$

Induct on $|2n - d|$, with this range statement as the IH at all smaller inputs with positive divisor.

- In either recursive row, the divisor stays $d > 0$ and the remainder is unchanged. Apply the IH to $(n + d, d)$ or $(n - d, d)$.
- If $n = d$, the remainder is 0, and $0 < d$.
- In the final row, the failed tests give $nd \geq 0$ and $d(n - d) \leq 0$. Since $d > 0$, these imply $0 \leq n \leq d$. The failed equality test gives $n \neq d$, hence $n < d$. Here $M(n, d) = n$.

The positive-divisor premise is used to cancel d without reversing inequalities.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.6, Examples 6.6.3–6.6.4.

Lean names and the existence theorem

The textbook calls M `fmod` and D `fdiv`. These are the relevant lemma applications:

```
-- Context: n d : Int, hd : 0 < d
have hid := fmod_add_fdiv n d
have hlo := fmod_nonneg_of_pos n hd
have hhi := fmod_lt_of_pos n hd
```

Together they give witnesses $q = D(n, d)$ and $r = M(n, d)$ for

$$\forall n, d \in \mathbb{Z}, \quad d > 0 \Longrightarrow \exists q, r \in \mathbb{Z}, \quad n = r + dq \ \wedge \ 0 \leq r < d.$$

Also $n - r = dq$, so $n \equiv r \pmod{d}$: every integer has a representative among $0, 1, \ldots, d - 1$.

The recursive proofs use `split_ifs`, `cancel`, and the source's `termination_by _ n d => 2 * n - d`; integer size means absolute value.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.6, Examples 6.6.2–6.6.5.

Practice 6.6: uniqueness and a negative input

1. Follow the recursive definition to compute $M(-8, 3)$ and $D(-8, 3)$. Record $|2n - 3|$ at each call.
2. Let $n, d, q, q', r, s \in \mathbb{Z}$ with $d > 0$ and

$$n = r + dq = s + dq', \qquad 0 \leq r < d, \quad 0 \leq s < d.$$

   Prove $r = s$ and then $q = q'$.

Hint: subtract the reconstruction equations. How large can $r - s$ be? Which integer multiples of d fit strictly between $-d$ and d?

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.6.6.

Solution 6.6: a bounded multiple must be zero

Computation: the inputs are $-8, -5, -2, 1$ with divisor 3. The measures are $19, 13, 7, 1$. At 1, the output is $(M, D) = (1, 0)$; three returns subtract 1 from D, giving

$$M(-8, 3) = 1, \qquad D(-8, 3) = -3, \qquad -8 = 1 + 3(-3).$$

Uniqueness: subtraction gives $r - s = d(q' - q)$, and the bounds imply $-d < r - s < d$. Put $t = q' - q \in \mathbb{Z}$.

- If $t \geq 1$, then $dt \geq d$, impossible.
- If $t \leq -1$, then $dt \leq -d$, impossible.

Thus $t = 0$, so $q' = q$ and $r = s$. The existence theorem plus this argument gives a unique quotient–remainder pair.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.6.6.

6.7 Euclid's algorithm, including negative inputs

For $a, b \in \mathbb{Z}$, define $G(a,b) = \gcd(a,b)$ recursively by

$$G(a,b) = \begin{cases} G(b, M(a,b)), & b > 0, \\ G(b, M(a,-b)), & b < 0, \\ a, & b = 0,\ a \geq 0, \\ -a, & b = 0,\ a < 0. \end{cases}$$

The measure is $|b|$. For $b \neq 0$, put $r = M(a,|b|)$. The remainder theorem gives $0 \leq r < |b|$, so $|r| < |b|$.

Nonnegativity proof: induct on $|b|$ with IH $G(x,y) \geq 0$ whenever $|y| < |b|$. If $b = 0$, $G(a,0) = |a| \geq 0$; otherwise $G(a,b) = G(b,r) \geq 0$ by the IH. In particular, $G(0,0) = 0$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.7, Definition 6.7.1; Example 6.7.2.

Why the output divides both inputs

Claim: $G(a,b) \mid a$ and $G(a,b) \mid b$ for all $a, b \in \mathbb{Z}$. Induct on $|b|$, assuming *both divisibility claims* for all smaller second-coordinate absolute values.
For $b \neq 0$, let $r = M(a,|b|)$, $q = D(a,|b|)$, $g = G(b,r)$. The IH gives $b = gu$ and $r = gv$ for some $u, v \in \mathbb{Z}$.

$$a = \begin{cases} r + bq = g(v + uq), & b > 0, \\ r - bq = g(v - uq), & b < 0. \end{cases}$$

Thus g divides a as well as b, and $g = G(a,b)$.
If $b = 0$, $G(a,0) = |a|$: use witness 1 for $a \geq 0$, witness -1 for $a < 0$, and witness 0 for divisibility of $b = 0$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.7, Example 6.7.3.

Extended Euclid: compute coefficients along with the gcd

Define $L, R : \mathbb{Z}^2 \to \mathbb{Z}$ mutually. For $b \neq 0$, put $r = M(a, |b|)$ and $q = D(a, |b|)$.

Condition	$L(a,b)$	$R(a,b)$
$b > 0$	$R(b,r)$	$L(b,r) - qR(b,r)$
$b < 0$	$R(b,r)$	$L(b,r) + qR(b,r)$
$b = 0,\ a \geq 0$	1	0
$b = 0,\ a < 0$	-1	0

Both functions only call the pair (b, r), whose measure $|r|$ is below $|b|$.
The intended invariant is

$$L(a,b)a + R(a,b)b = G(a,b).$$

The change of sign in R matches $a = r + bq$ versus $a = r - bq$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.7, Definition 6.7.4.

Worked computation: descend, then recover coefficients

For $(a, b) = (-26, 18)$, divide with positive divisors:

$$-26 = 10 + 18(-2), \quad 18 = 8 + 10(1), \quad 10 = 2 + 8(1), \quad 8 = 0 + 2(4).$$

The second inputs have measures $18, 10, 8, 2, 0$, so the gcd is 2. Recover (L, R) from the last row upward:

(a, b)	q	$L(a, b)$	$R(a, b)$
$(2, 0)$	—	1	0
$(8, 2)$	4	0	1
$(10, 8)$	1	1	-1
$(18, 10)$	1	-1	2
$(-26, 18)$	-2	2	3

Check: $2(-26) + 3(18) = 2$. For example, the last coefficient is $-1 - (-2) \cdot 2 = 3$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.7, Definition 6.7.4 (numerical variant).

Proving the coefficient identity: positive second input

Theorem: $L(a,b)a + R(a,b)b = G(a,b)$ for all $a, b \in \mathbb{Z}$. Induct on $|b|$. The IH is

$$L(x,y)x + R(x,y)y = G(x,y) \quad \text{whenever } |y| < |b|.$$

If $b > 0$, let $r = M(a,b)$ and $q = D(a,b)$, so $a = r + bq$. Write $\ell = L(b,r)$ and $t = R(b,r)$. The IH gives $\ell b + tr = G(b,r)$. Therefore

$$\begin{aligned} L(a,b)a + R(a,b)b &= ta + (\ell - qt)b \\ &= t(r + bq) + (\ell - qt)b \\ &= tr + \ell b = G(b,r) = G(a,b). \end{aligned}$$

The cancellation of tbq explains the definition of $R(a,b)$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.7, Example 6.7.5.

The negative branch and the two terminal cases

If $b < 0$, let $r = M(a, -b)$ and $q = D(a, -b)$. Now $a = r - bq$. Put $\ell = L(b, r)$ and $t = R(b, r)$. The IH still says $\ell b + tr = G(b, r)$, since $|r| < |b|$.

$$\begin{aligned} L(a,b)a + R(a,b)b &= ta + (\ell + qt)b \\ &= t(r - bq) + (\ell + qt)b \\ &= tr + \ell b = G(b, r) = G(a, b). \end{aligned}$$

For $b = 0$ there is no recursive call:

$$\begin{aligned} a \geq 0: &\quad 1 \cdot a + 0 \cdot b = a = G(a, 0), \\ a < 0: &\quad (-1) \cdot a + 0 \cdot b = -a = G(a, 0). \end{aligned}$$

These cases complete the induction for all integer signs.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.7, Example 6.7.5.

Lean: mutual recursion and a reusable existence theorem

The source defines `L` and `R` in a `mutual` block. Its `gcd_dvd_left` and `gcd_dvd_right` proofs also use mutual induction. All recursive calls decrease $|b|$.

Bézout's identity: for every $a, b \in \mathbb{Z}$, there are $x, y \in \mathbb{Z}$ with $xa + yb = G(a, b)$. Choose $x = L(a, b)$, $y = R(a, b)$.

```
-- In the source's Euclid namespace:
example (a b : Int) :
    Exists (fun x : Int => Exists (fun y : Int =>
      x * a + y * b = gcd a b)) := by
  refine Exists.intro (L a b) ?_
  refine Exists.intro (R a b) ?_
  exact L_mul_add_R_mul a b
```

The witnesses need not be unique: replacing (x, y) by $(x + tb, y - ta)$ for $t \in \mathbb{Z}$ preserves the equation.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.7, Examples 6.7.3–6.7.6.

Practice 6.7: certify the gcd

1. Compute $G(14, -8)$, $L(14, -8)$, and $R(14, -8)$. Start with $14 = 6 + 8 \cdot 1$. Apply the negative-b branch at the first step. Verify the resulting linear combination.
2. Let $a, b, d \in \mathbb{Z}$. Prove

$$d \mid a \ \wedge \ d \mid b \quad \Longrightarrow \quad d \mid G(a, b).$$

   Use Bézout's identity and divisibility witnesses, without a new induction.

Explain why this property, common divisibility, and nonnegativity characterize the gcd even when a or b is negative.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.7.7; Definition 6.7.4.

Solution 6.7: coefficients certify maximality

Computation. The recursive pairs are

$$(14, -8) \to (-8, 6) \to (6, 4) \to (4, 2) \to (2, 0).$$

The quotients (using $|b|$) are $1, -2, 1, 2$. Backward coefficient pairs are

$$(1, 0),\ (0, 1),\ (1, -1),\ (-1, -1),\ (-1, -2).$$

Thus $G(14, -8) = 2$ and $(-1)14 + (-2)(-8) = 2$.

Divisibility. Write $a = du$, $b = dv$ and choose Bézout witnesses x, y. Then

$$G(a, b) = xa + yb = xdu + ydv = d(xu + yv).$$

If $(a, b) \neq (0, 0)$, the gcd is positive, so every positive common divisor is at most it. At $(0, 0)$ the gcd is 0; the divisibility characterization still holds, although there is no largest positive common divisor.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.7.7.

References

Heather Macbeth, *The Mechanics of Proof*, Chapter 6, "Induction."

- Sections 6.1–6.3: introduction, recurrence relations, two-step induction.
- Sections 6.4–6.5: strong induction and Pascal's triangle.
- Sections 6.6–6.7: the Division Algorithm and the Euclidean algorithm, including the extended algorithm and Bézout's identity.

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